

SARTHAK ANAND

Expertise on domains – evaluation harnesses, multi-turn dialogue agents, RAG pipelines, and prompt/inference optimisation in production. Prior depth in **model compression** and **multi-modal learning**. Research-first: reproduce and stress-test published methods before adopting them; own the problem end to end, from framing the metric to defending the number.

Education

Degree	Institute	Score	Year
B.Tech. Major (Civil)	Indian Institute of Technology (BHU), Varanasi	9.72	2025
B.Tech. Minor (CSE)	Indian Institute of Technology (BHU), Varanasi	9.18	2025
Senior Secondary	Delhi Public School, Ruby Park, Kolkata (CBSE)	96.4	2021
Secondary	Delhi Public School, Ruby Park, Kolkata (CBSE)	96.0	2019

Experience

Machine Learning Engineer | Convin.ai

June 2025 – Present

Text-to-SQL Agent over Enterprise Databases

- Architected a **multi-hop agent** over **5** databases (**45** tables, **~86** columns, **500K** rows) - intake → classify_intent → discover_schema → resolve_values → supervisor; **46%** first-turn, **86%** second-turn, **95%** user satisfaction.
- Unified **KG** of the ERD in **Neo4j**; vector search + shortest-path to construct the **min. distinguishing schema**.
- N-candidate** SQL from divergent reasoning strategies & parallel cross-verification; **oracle selection** over the set.
- GRPO**-finetuned SQL verifier on **retrieved columns**, reward **1** execution-result match, **0.1** valid-but-wrong SQL, **0** malformed. Training set curated by **k-sample difficulty** filter, dropping **0/k** and **k/k** questions.
- Whole fan-out affordable on **1.7B** models against a **32B** reasoning supervisor; **speculative decoding** for latency.
- Read-only by design with per-user-group encryption policy; drift monitoring diffing the knowledge graph against live catalog metadata; business rules and per-schema query conventions in persistent cross-session memory.

Indic Book Translation — Multi-Stage Correction Pipeline

- Gemma** harness over **18** Indian languages incl. **low-resource** ones; **phrase-level-trained** draft model behind an **[N]-sentinel** block codec — script-agnostic anchors surviving Indic tokenization, whole page in **1** call at **1:1** alignment.
- Architected the error corrector** as a **10-node LangGraph** loop — parallel **script-purity** + **terminology** checkers → aggregate → propose → verify → apply; **propose** split from **verify** so no edit lands unreviewed; bounded.
- PPO-trained** corrector on a hybrid reward - **ORM**: error set empties; **PRM**: per-edit verified · applied · no-new-error, plus a step penalty for **turn-level credit assignment** — **4.4** → **3.1** passes to convergence.
- Hallucination bounded structurally** — model emits **span-level** edits; deterministic **3-tier splice** (exact, regex, fuzzy) over **NFC-normalized** text; step drafting fallback chain (block shuffle, per-block, context-surround).
- Global page intelligence** - a **librarian** pass once per page and conditions every block; **glossary** of translations persisted across page and chapter, holding terminology stable over **22** chapters.
- Human eval feeding an **RLHF** loop and self-evolving correction rules; **2×H200** serving **~5 s/page**, reviewers scored output **9/10** and **above** the incumbent NCERT translation; won a **Ministry of Education (AICTE)** tender.

Yaksha — Omnichannel Conversation Intelligence

- Architected a two-service Go platform** (control plane + deliberative engine) via a unified gRPC contract. Engineered a bidirectional Turn stream with **deferred state commit** – ensuring perfect LLM-context-to-audio alignment.
- Solved DB pool starvation** during multi-second LLM calls by scaling state behind Postgres via optimistic concurrency (rev tripwires on versioned JSON blobs). Decoupled I/O via injectable seams for hermetic, sub-second core test suites.
- Engineered a bounded deliberative pipeline** featuring a hybrid retrieval module and a global slot-belief store to prevent redundant questioning.
- Constructed a dynamic conversational topic graph** bootstrapped from live human transcripts. Implemented an ingestion loop to monitor live conversations and dynamic node creation for out-of-distribution flows.
- Built the real-time voice path** (WebRTC → STT → FSM → TTS) using chunking for immediate streaming.
- Shipped an idempotent orchestration loop** for proactive next-best-action outreach, featuring burst debouncing, channel-specific cooldown anchors, and dynamic DND time-shifting with daily spread caps.

LLM Agent Eval & Self-Optimization Harness

- Agent eval harness** replaying a production — **persona-policy** user sim with adversarial commitment (non-agreeable by construction), RAG tool-call action loop, **LLM-as-judge** rubric scoring; pass rate **24%**→**76%** on a **[45]-case** bank.
- Atomic binary criteria** — structural verification where checkable and LLM adjudication elsewhere. **3-judge** panel (**Qwen**, **gpt-oss**, **Gemma**) scoring **raw traces**; **16** passes/trial fixing a **±0.14** noise floor - no edit called a win beneath.
- Optimization on a **Pareto front** of weighted score × **turns-to-resolution** — **0.21/18** → **0.67/10**, a **3×** noise margin.
- Split a **frozen regression rubric** from a **discovery rubric** mined from live traces — avoiding the **reward misalignment**. Each prompt edit **pre-registered** with predicted fix and predicted regression, then measured.

Embedding-Space Engineering for Conversation Taxonomy

- Diagnosed **mmBERT-base** embeddings with a 7-metric geometry suite + cross-lingual **FLORES** alignment scoring.

- Fine-tuned encoder with **supervised contrastive loss** + hard-negative mining on language-stratified splits (also benchmarked **triplet loss**), closing a **0.9996** AUC leak — separation margin **0/23→23/23** (held-out **+0.96**).
- Added a **relational distillation** term (frozen-teacher pairwise-similarity penalty) to recover alignment against separation.
- Trained **MLP head** on frozen embeddings; joint encoder+head fine-tuning lifted accuracy & cut down FPR and FNR.
- Shipped into a real-time **Go** voicemail-detection service — reorder buffer for async per-chunk scores, cost-asymmetric decisioning; validated at **89%** precision / **93%** recall (**AUC 0.98**) vs. **59%** precision for the **GPT-4o-mini** baseline.
- Served embeddings via **TEI** (Rust) on a single A10 — switched to a **MiniLM** backbone (fp16) for **28.6K texts/s** peak throughput and a **p99 41ms at 19.2K texts/s**, $\sim 22\times$ mmBERT's ceiling.

Model Adaptation & Inference Efficiency

- **SFT** of **Qwen** and **Gemma** models on consumer conversation data for **NER** and AI-insight extraction.
- Restructured prompt templates for **KV-cache**-friendly prefixes; per-role **prompt_cache_key** routing on an enterprise endpoint with no native cache guarantee — **61%** of a live production call measured cacheable, **~30% projected** input-cost reduction.

Summer Intern | JSW One Platforms

May 2024 – July 2024

- **Vendor–client matchmaking** and **regional risk grading** workflow with KPI-based vendor feature engineering.
- Sentiment analysis on free-text feedback; **PCA** and **UMAP** reduction to an **8-feature** signal set.
- Probabilistic loss in a **ListNet LTR** ranker — **NDCG 0.85**; scores surfaced via the **Salesforce/PowerBI** stack.

Research Intern | Centre for Sustainable Technologies, IISc Bangalore

May 2023 – July 2023

- Tri-directional strength of **3D-printed concrete** across chemical, rheological and structural composition combinations.
- **LSTM** trained over **40 FEM** models for real-time stress distribution on cured layers — **MAE 0.43**, **0.6 s** runtime.
- **Random Forest** benchmarked best against DT and SVR — R^2 **0.97**, **MAE 4**.

Tech: Python, Go, asyncio, PyTorch, DSPy, LLM-as-Judge, RAG, Knowledge Graphs, Vector Search, PPO / RLHF, SFT / PEFT, On-Policy Distillation, Speculative & Graph Decoding, KV-Cache Optimisation, Qwen / Gemma / Llama, Embeddings & Whitening, SQL, ListNet / Learning to Rank, PCA, UMAP, Salesforce / PowerBI, LSTM, Random Forest, SVR, FEM.

Projects

- **Amex Campus Challenge — Profitability Formula Reverse-Engineering** (Jul 2026) — **Inverse-optimization** pipeline recovering a hidden profit-ranking formula from a **10-query** black-box leaderboard (500K members, no labels) via multi-start fits and cluster-medoid selection — **0.795** top-20% overlap; self-directed oracle rebuild proved the formula **under-identified** and an imputation heuristic **5–8 $\times$** miscalibrated.
- **Multimodal Entity Extraction** (Sep 2024) — **OCR→prompt-tuned Llama 3.1-Instruct 7B** pipeline over regex-normalised spans, benchmarked against direct **VLM** extraction (Qwen-VL 2B/7B, ViT) for product-attribute parsing — **F1 0.67**.
- **Filter Pruning in ResNet-50 / VGG-19** (Sep – Nov 2023) — Knee-guided **genetic search** over binary-genome filter masks along the complexity–accuracy **Pareto front**, fitness evaluated in **parallel** on HPC — parameters **–78%** / **–82%** at **–2%** / **–4%** accuracy.
- **Multi-Modal Stock Prediction** (Jan – Mar 2024) — Dual-branch **BiLSTM** on **RSI**-engineered S&P 500 series fused with **fine-tuned FinBERT** three-class sentiment over scraped news — **74.6%** accuracy vs. **57.4%** time-series only.
- **Integer-bit Quantization in DNNs** (Jul – Aug 2023) — **PTQ** implementations compared across PyTorch and TensorFlow with multi-threaded multi-core GPU execution — reduced weight-matrix storage at higher throughput.
- **Signal-Processing ML for Structural Health Monitoring** (2023 – 24) — **SSA + AMUSE** blind source separation recovering modal frequencies and mode shapes under heavy noise, with a **sparse autoencoder** quantifying damage location and extent — **MAC 0.971**, **MSE 8.2e-05**; plus a **Gaussian-process** digital twin of linear oscillators. Technical paper drafted.

Tech: Inverse Optimization, Coordinate Descent, Identifiability Analysis, PyTorch, TensorFlow, Keras, HuggingFace Transformers, DocTR / EasyOCR, VLMs, Prompt Tuning, BERT, Model Compression, PTQ, CUDA, Distributed Training, Web Scraping, CIFAR-10, MATLAB, BSS, GP Regression, Monte Carlo.

Honors and Awards

- **Department Rank 1**, Civil Engineering, IIT (BHU), 2025 — CRS Iyengar Memorial Gold Medal, Rai Bahadur Taracharan Gue Memorial Award, Late Prof. Manoranjan Sengupta Platinum Jubilee Merit Award, IIT (BHU) Medal.
- **OPJEMS Scholarship** 2023 & 2024, Best Business Idea Proposal — *Damage Mitigation through WSN*.
- **Hyundai Hope Scholarship** for aspiring innovators from IITs — *Formula Student Racing Car Design*.
- Qualified, **L&T Createch 2024** — *Predictive Modelling for Agile Workflow*.
- College Team Captain, Collegiate Chess League 2024; Gold Medal, Chess Meet, IISc Bangalore.